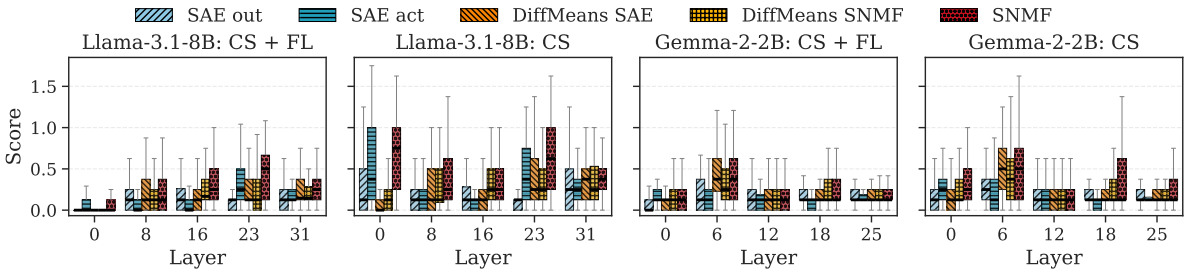

Figure 3: Causality evaluation results across layers in Llama-3.1-8B and Gemma-2-2B, using publicly-available SAEs (`SAE out`), SAE trained on MLP activations (`SAE act`), `SNMF`, and `DiffMeans` trained on concepts detected by SAE or `SNMF`. Concept Steering + Fluency (CS + FL) measure steering performance while preserving coherence, and Concept Steering (CS) quantifies how strongly the target concept is steered. Across both models `SNMF` scores often exceed both `DiffMeans` and SAEs, showing strong causal influence.

layers tend to show lower final scores but relatively high concept scores. This discrepancy arises from the sensitivity of early-layer steering. Modifications at these layers propagate through the network, often strongly influencing model fluency and as such impacting the final score.

Notably, `SNMF` consistently outperforms SAEs across all layers and often matches or exceeds the performance of `DiffMeans`, a supervised baseline. Since `DiffMeans` computes feature directions via subtraction of positive and negative representations, it is susceptible to noise by unrelated concepts present in the positive samples. This issue is especially prominent in earlier layers, where limited contextualization leaves token representations within the positive sample distinct, making their average less meaningful. In contrast, `SNMF` produces a parts-based decomposition that captures structure consistently associated with the target concept, leading to improved performance across all layers.

These results demonstrate that `SNMF` is able to isolate groups of neurons encoding concepts, supporting its viability as an interpretable unsupervised method for discovering meaningful directions through the MLP. Furthermore, the ability of MLP features produced by `SNMF` to reliably steer model outputs implies that the MLP operates using additive updates composed of interpretable neuron sets, which extends the findings by Geva et al. (2022) that single MLP vectors promote concepts in the residual stream. Each MLP feature corresponds to a distinct combination of neurons whose activations affect the residual stream in ways aligned with human-interpretable concepts. Considering that neuron combinations activate to represent a concept, it follows naturally that a single neuron may participate in multiple concepts, and that a single concept may activate multiple neurons.

6 Analyzing Neuron Compositionality

Our experiments show that `SNMF` finds groups of neurons that represent interpretable features. A natural question that arises is *how are different neurons combined to form features?* Here, we investigate this question using the properties of `SNMF`.

Feature Merging via Recursive SNMF We apply `SNMF` recursively with progressively smaller values of k, which encourage more generic patterns (Song and Lee, 2013). Namely, we start by decomposing the activations and then recursively decompose the generated MLP features. This process reveals subsets of neurons that are shared across features, forming a hierarchy that shows how the model additively combines groups of neurons to represent concepts. We run this experiment on a dataset of sentences involving the concept of time units (e.g., months, weekdays, minutes) and use the coefficient matrices Y from each step to identify relationships between MLP features. More details on the recursive application of `SNMF` are in §F.

Figure 4 illustrates the resulting hierarchy of concepts corresponding to the MLP features learned at each step. Concept labels were derived from the top activating contexts of the MLP feature according to the coefficient matrix. For example, the *weekend* node was activated for contexts mentioning Saturday, Sunday and the word "weekend". Overall, we see that the resulting structure progresses from groups of neurons that represent fine-grained concepts (i.e., specific weekdays) to more high-level concepts (middle/end of the week and "day of week"). This feature-merging behavior is the inverse of what has been described as feature splitting in SAEs (Chanin et al., 2024; Bricken et al.,

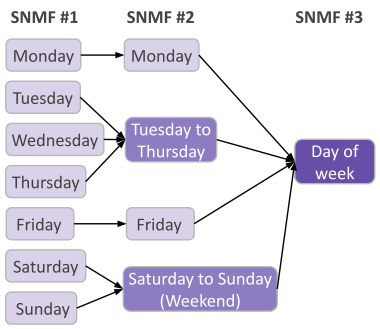

Figure 4: Example of feature merging via recursive SNMF, showing the concepts corresponding to the features identified at each step. Features were obtained for activations in GPT-2 Small on the weekdays dataset.

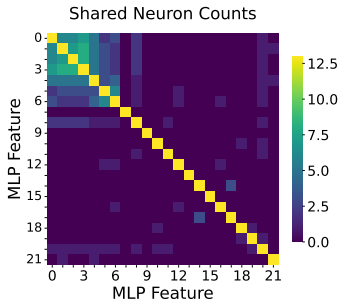

Figure 5: Shared neuron counts between MLP feature pairs. Features 0–6 correspond to weekdays (Monday-Sunday). the rest are randomly sampled from Z. All days share core neurons, suggesting a general day representation, while weekdays and weekends share distinct subsets, forming specialized neuron cores.

2023), where a single feature in a smaller SAE corresponds to multiple features in a larger SAE. Additional recursive decomposition examples learned from a more general dataset are provided in §F.

Semantically-Related MLP Features Share Common Neuron Structures The previous analysis suggests that certain groups of neurons contribute to semantically-related MLP features. Here, we look into these shared structures using the matrix Z. We binarize Z with a per-feature threshold to produce $\bar{Z}$, where the top activating neurons of each MLP feature are set to 1 and the rest to 0. Then, we calculate $M := \bar{Z}\bar{Z}^T$, where each entry $M_{i,j}$ is the number of overlapping, top activating neurons between the MLP features i and j.

Figure 5 visualizes M, where the first seven rows correspond to MLP features associated with Monday through Sunday and the rest MLP features are randomly sampled from Z. The heatmap reveals that the model represents the concept of weekdays using a common set of neurons, as indicated by the higher shared neuron counts among the first seven rows and columns. Given the increased diagonal values, we understand that each day also activates its own additional distinct subset of neurons. Furthermore, the MLP features representing the weekend and those representing weekdays each share more neurons within their respective groups. We can see this in rows 5 and 6, which have higher shared neuron counts with each other than in rows 0 to 4 (and vice versa). Meanwhile, there is little overlap with the remaining MLP features. These patterns show that the model utilizes sets of neurons as building blocks to represent complex and hierarchical concepts. Specifically, a core set of neurons encodes the concept of a day, while additional neurons refine the representation into weekend or weekday, and further neurons produce representations for individual days.

Neuron group	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Monday	**2.0**	-0.8	-1.0	-1.2	-0.8	-1.1	-0.1
Tuesday	-3.3	**0.5**	-2.2	-2.4	-2.4	-3.2	-2.8
Wednesday	-2.7	-3.2	**2.5**	-2.7	-4.3	-3.2	-2.7
Thursday	-0.9	-1.0	**0.6**	**4.5**	-0.1	-1.7	-0.5
Friday	-2.9	-2.8	-2.7	-1.6	**1.3**	-1.9	-2.6
Saturday	-0.2	-0.7	-0.1	-0.7	-0.5	**2.4**	**0.7**
Sunday	-0.4	-1.7	-1.6	-2.0	-0.8	-0.8	**2.6**
Core weekday	**5.8**	**5.7**	**5.4**	**5.8**	**6.0**	**5.4**	**4.7**

Table 2: Change in logits for weekday tokens (columns) when amplifying exclusive neurons tied to different weekday concepts (rows). Positive values (black) indicate promotion; negative (red) indicate suppression. Exclusive neurons promote their own token (diagonal) and suppress others, while core neurons (bottom row) raise all logits, encoding a general "weekday" concept.

We test this via causal interventions. Let the *neuron base* be the neurons shared across all weekday MLP features, and the *exclusive neurons* the unique neurons to each MLP feature. We feed the prompt "`I think that`" to GPT-2 Large, while amplifying the neuron base and, separately, the exclusive neurons of each weekday. We measure the effect of each intervention on the model's output distribution by calculating the change in logits for tokens associated with each day of the week.

Table 2 shows that amplifying the neuron base promotes all weekday tokens, suggesting it encodes a high-level weekday concept. Conversely, amplifying a day's exclusive neurons promotes its associated token while suppressing the others. For example, activating Sunday's exclusive neurons

suppresses the tokens for "Monday" through "Saturday". These findings demonstrate that the model builds increasingly specific concepts by using sets of neurons as compositional building blocks.

7 Conclusion

We introduce an unsupervised method to decompose MLP activations into MLP features, which represent how groups of neurons contribute to encoding concepts. The MLP features often align with human-interpretable concepts, performing on par with SAEs on interpretability benchmarks, while achieving better steering performance than both SAEs and a strong supervised baseline, `DiffMeans`. We further demonstrate that the model composes groups of neurons additively to represent a wide range of concepts, revealing a hierarchical organization in the hidden activation space. This behavior helps explain why the same sets of neurons contribute to multiple features, extending the functional role of neurons in the MLP. It also supports the view that the feature splitting observed in SAEs reflects the model's use of neuron composition to construct more specific features, rather than necessarily being a byproduct of SAE training. Overall, our findings provide a new lens on how MLPs form features in LLMs and we introduce an interpretable unsupervised approach for identifying and interpreting MLP features in transformers.

Limitations

While our work introduces a promising method for analyzing MLP activations, our benchmarks were limited to number of MLP features of $k < 500$. Analyzing the granularity of features that may emerge at larger values of k and whether the method scales effectively to thousands of MLP features are interesting directions for future work.

In terms of optimization of `SNMF`, we did not perform a rigorous comparison of optimization methods or initialization strategies. We chose sampling-based initializations (uniform and normal) for simplicity. However, given that `SNMF` is non-convex, different initializations of the matrices Y and Z may converge to different local minima. Previous papers have suggested initializations such as K-means (Ding et al., 2010) or SVD based (Boutsidis and Gallopoulos, 2008) that may provide improved concept detection and steering. Moreover, using projected gradient descent rather than Multiplicative Updates may allow for regularization, further improving performance.

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